

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	28 June 2026
Team ID	
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Strategic Product Placement Analysis using Machine Learning and Data Analytics

Reference: <https://cloud.google.com/architecture>

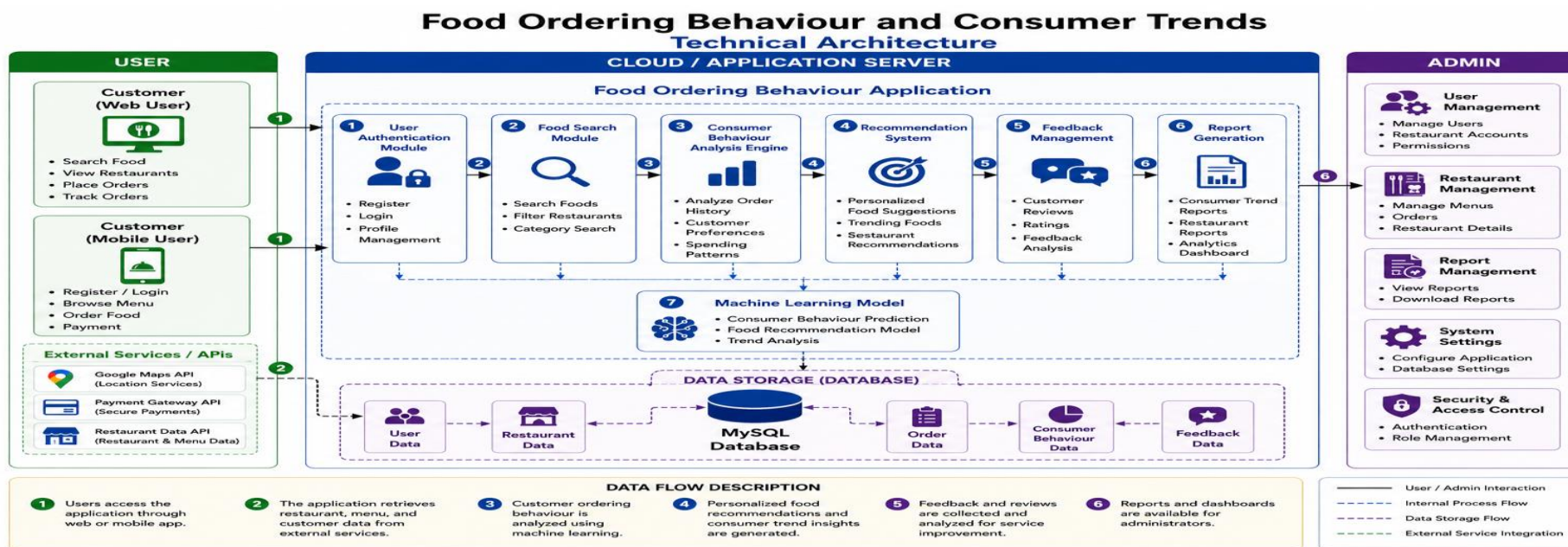


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web application interface for customers and administrators to interact with the system.	HTML, CSS, JavaScript, Bootstrap
2	Application Logic-1	User registration, login, authentication, and profile management.	Python, Flask
3	Application Logic-2	Food search, restaurant browsing, order processing, and consumer behaviour analysis.	Python, Pandas
4	Application Logic-3	Food recommendation generation, trend analysis, report generation, and dashboard visualization.	Scikit-learn, Matplotlib
5	Database	Stores customer information, restaurant details, food menus, orders, consumer behaviour, and reports.	MySQL
6	Cloud Database	Cloud-based storage and backup of application data.	Firebase / MySQL Cloud
7	File Storage	Stores generated reports, datasets, exported files, and analytics results.	Local File System, Google Drive
8	External API-1	Retrieves restaurant locations and map information.	Google Maps API
9	External API-2	Online payment processing and transaction verification.	Payment Gateway API (Stripe/Razorpay)

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for frontend and backend development.	HTML, CSS, Bootstrap, Flask
2	Security Implementations	Password encryption, authentication, secure access control, and data protection.	SHA-256, HTTPS, Flask Authentication
3	Scalable Architecture	Three-tier architecture supports increasing users, restaurants, and orders.	Three-Tier Architecture
4	Availability	System remains available through cloud hosting and automated data backup.	Cloud Hosting, Backup Services

References:

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

<https://flask.palletsprojects.com/>

<https://cloud.google.com/architecture>

